Branch-point norm orders: independent derivation

Mathematical source review

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Derived distinction. At a simple rational branch point, the curve-stalk order of the norm is twice the order of the selected numerator. The base-polynomial order of that same norm equals the numerator order. These are different valuations.

Setting. Let K be a field of characteristic different from two, $f \in K[x]$ a separable quintic, $P = (a, 0)$ with $f(a) = 0$, and $R = \mathcal{O}_{C,P}$ for $C : y^2 = f(x)$. Write v_P for its normalized DVR order and v_a for polynomial multiplicity at a. Let $b \in K^\times$, $n \in K[x]$, $h = by - n(x)$ and $N = n^2 - b^2 f$.

Local geometry. Write $f = (x - a)g$. Since $g(a) = f'(a) \neq 0$, g is a local unit and $x - a = y^2/g$. The maximal ideal is (y); the actual smooth curve local ring is a DVR. Hence

$$v_P(y) = 1, \qquad v_P(x - a) = 2, \qquad v_P(A(x)) = 2v_a(A).$$

If $n(a) \neq 0$, h is a unit. If $n(a) = 0$, write $n = (x - a)k$ and obtain $h = y(b - yk/g)$, with unit second factor. Its conjugate is $y(-b - yk/g)$, also of order one. Thus

$$v_P(h) = v_a(N) \in \{0, 1\}, \qquad v_P(N) = 2v_P(h).$$

The norm is nonzero: at a selected branch its base order is one because n^2 has order at least two while $b^2 f$ has order one; otherwise its residue is $n(a)^2 \neq 0$.

Factored source norm. If the exact identity is $N = cu_1u_2U$ with $c \in K^\times$, all factors are nonzero and

$$v_a(u_1) + v_a(u_2) + v_a(U) = v_P(h),$$
$$v_P(u_1(x)) + v_P(u_2(x)) + v_P(U(x)) = 2v_P(h).$$

At a selected branch these sums are one and two respectively. Exactly one factor has a simple base root; simultaneous roots of two factors contradict the premises. The sign convention $n - by$ changes no order. A source support-root condition implies $n(a) = 0$ by evaluating the norm identity.

Worked counterexample. Over $\mathbf{Q}$, take $f = x^5 + x$, $a = 0$, $b = 1$, $n = x$. The polynomial is separable: at a nonzero root $\alpha^4 = -1$, its derivative is -4; at zero it is one. Here

$$h = y - x, \qquad N = x^2 - x^5 - x = -x(1 - x + x^4).$$

Thus $v_P(h) = 1$, $v_a(N) = 1$, and $v_P(N) = 2$. Both transplanting the nonbranch formula $v_P(h) = v_P(N)$ and blindly asserting $v_P(h) = 2v_a(N)$ would give $1 = 2$. This is a general polynomial-factor example, not a claimed coefficientData fixture.

Premise attacks. If $b = 0$ and $n = x - a$, numerator order is two: the zero-or-one bound needs $b \neq 0$. Repeated roots can destroy the DVR/uniformizer argument. Characteristic two changes the distinct-conjugate interpretation, although some elementary local identities can survive. Infinity and nonrational closed points require separate definitions and proofs.

Primary source correspondence. Existing WeightedProjFiniteBranchDVR constructs the actual branch stalk, DVR and orders of y and $x-a$. WeightedProjSourceBranchActualQuintic proves source numerator order one under determinant, input-congruence and support-root premises. Exact source hashes are retained in derivation.json. No new producer candidate was read; no compiler or trial was run. This report is a hand derivation and source inspection, not fresh kernel evidence or a global divisor-class result.